

RHESSys Manual

(under development)

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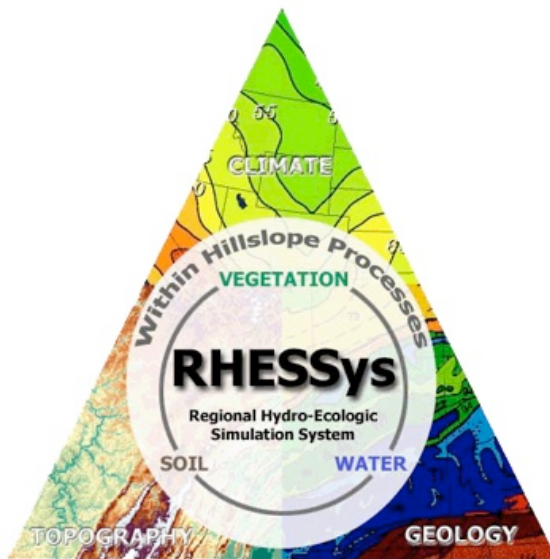
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1 Welcome to the RHESSys Manual

(under development)



[Tague Team Lab](#) | [GitHub](#)

1.1 Purpose

Include info about the manual, why it exists, what it is intended for

1.2 Scope

- What to expect from this book (user will learn....)
- What the manual covers (practical step-by-step guidance)
- What it does not cover (underlying theory of the model, all possible functionality)

1.3 Intended Audience

- Who this is for/Who should be reading this

1.4 Assumed basic prerequisite knowledge

- General Modeling
- Basic R/Rstudio
- GIS and Raster/Vector
- Basic Unix/Linux environment (shell, command line, text editor)
- Git/GitHub

2 About this Manual

2.1 Conventions

- How to read this document
- Key/Legend for document formatting - regarding use of icons, symbols, different font indications

2.2 Manual Structure

- organization
- navigation
- links to sections
- potentially a link to a quick start guide

2.3 Support/Resources

- Potentially include author info (Naomi, Janet (contact info?), Ryan)
- RHESSys GitHub repositories (RHESSys code, RHESSysPreProcessing, RHESSys-IOinR, ParameterLibrary)
- RHESSys Wiki(s)
- Potentially a FAQ page, a user community forum (i.e. ‘issues’ section on RHESSys Github repo?)

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Part I

Part I: Foundations & Environment

4 Part I: Foundations & Environment

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- This part is about understanding the logic of this tool and setting the computational environment
- intro to what is covered in the section
- expected outcome for this section: functioning modeling environment and clear strategic plan

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7 1 Introduction (The Model)

7.1 1.1 Overview of RHESys

7.1.1 What RHESys is

- Description

7.1.1.1 Capabilities

- Integrated Ecohydrological Modeling
- Spatial distribution of the landscape
- Feedback loops
- Fire model

7.1.1.2 Limitations

- High data demands
- Can be computationally intensive
- Maybe any processes that are not well represented? Let users know if the model won't fit their research question

7.2 1.2 Recommended reading/Publications

- Foundational papers
- Overview/Review papers
- Applications

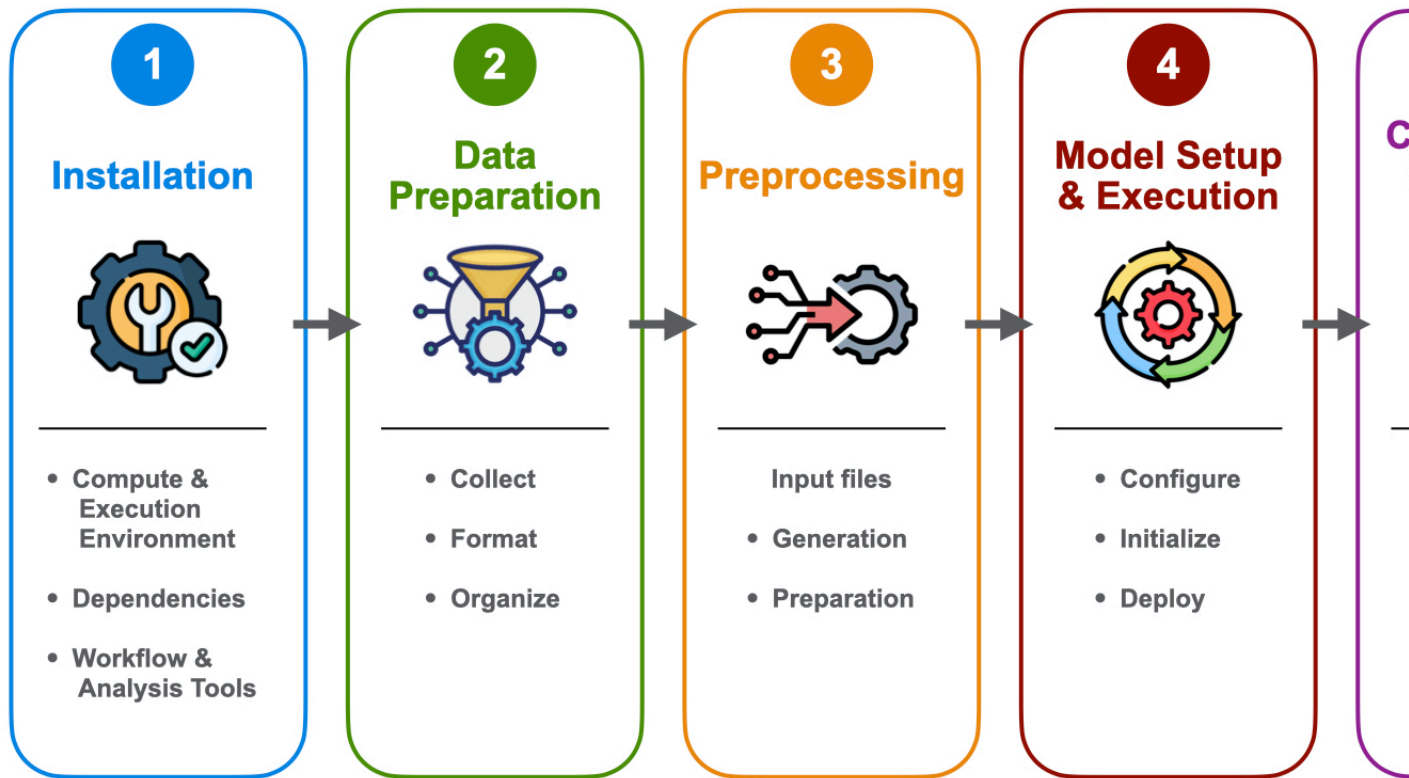


Figure 7.1: RHESys Modeling Workflow Overview

7.3 1.3 Workflow overview

- Step 1: Installation - Set up the RHESys environment
- Step 2: Data Preparation - Gather and format spatial, climate, and validation data
- Step 3: Preprocessing - Generate the landscape representation (Worldfile) and routing (Flowtable) input files
- Step 4: Model Setup and Execution - Define the simulation rules and run RHESys
- Step 5: Calibration & Sensitivity Analysis - Condition the model (Initialize/reach equilibrium), refine parameters, validate performance
- Step 6: Post Processing - wrangle, analyze, and visualize output data

7.3.1 Key concepts and terminology

(define frequently used terms)

- Spatial hierarchy (basin, hillslope, zone, patch, stratum)
- Input File types (Worldfile, Header, Flowtable, Tecfile, Parameter/Climate files, Output filters)
- State variables (store quantities) and fluxes
- Spatial/Temporal step (i.e., basin daily, patch monthly)

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9 2 Setup (The Software)

9.1 2.1 Setup options

(description of each, who should use, when to use, recommendation, link to corresponding install instructions)

9.1.1 2.1.1 Local Installation (MacOS, Linux)

- more advanced users
- users who want more flexibility

9.1.2 2.1.2 Containerized setup (Docker)

- beginners
- Windows users
- portability

9.1.3 2.1.3 HPC/Cloud (SIF/Singularity)

- high-performance computing environments
- cluster users
- large scale, high resolution simulations

9.2 2.2 RHESSys Installation

9.2.1 2.2.1 MacOS (include section for Linux as well if necessary or where differs)

9.2.1.1 Code Dependencies and Operational Requirements

- Terminal application
- Xcode Command Line Tools (compilers/build tools)
- Privileges (i.e. Administrator access)

- Homebrew
- Git
- Libraries (i.e. NetCDF)

9.2.1.2 Installation steps

- Download code from GitHub
 - Zip file
 - Clone repository (main, develop)
- Compile
- Shell configuration (adding to path)
- Verify installation
- Common issue and solutions

9.2.2 2.2.2 Docker

- Install Docker Desktop
- Obtaining RHESSys image from the Docker Hub

9.2.3 2.2.3 SIF

- Obtaining RHESSys .sif image

9.3 2.3 Installing Workflow & Analysis Tools

9.3.1 2.3.1 R Ecosystem

- R (The engine)
- RStudio (The interface)
- Standard Packages (Extensions - Libraries that expand functionality)

9.3.2 2.3.2 WhiteboxTools for Geospatial Processing

9.3.3 2.3.3 RHESSys-Specific R packages

- RHESSysPreprocessing
- RHESSysIOinR

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11 3 Project Design & Planning (The Strategy)

- will discuss what a user should consider, i.e.:

11.1 3.1 Spatial resolution vs. computational cost

- patch resolution/number of patches impact on processing needs & time
- scale - what is necessary to answer users questions: is fine-scale processing necessary?

11.2 3.2 Data synchronization & temporal extent

- overlap - i.e., does the temporal period of climate data & observed streamflow data coincide?
- spin-up and calibration time requirements

11.3 3.3 Functional constraints

- Feature specific inputs - i.e. to run the fire model, is wind speed/direction and relative humidity data available?
- Does RHESys output what is needed to answer research questions?

11.4 3.4 Defining the project boundaries

- use full basin boundary, representative hillslope, single patch?

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13 4 Example Dataset (The Sandbox)

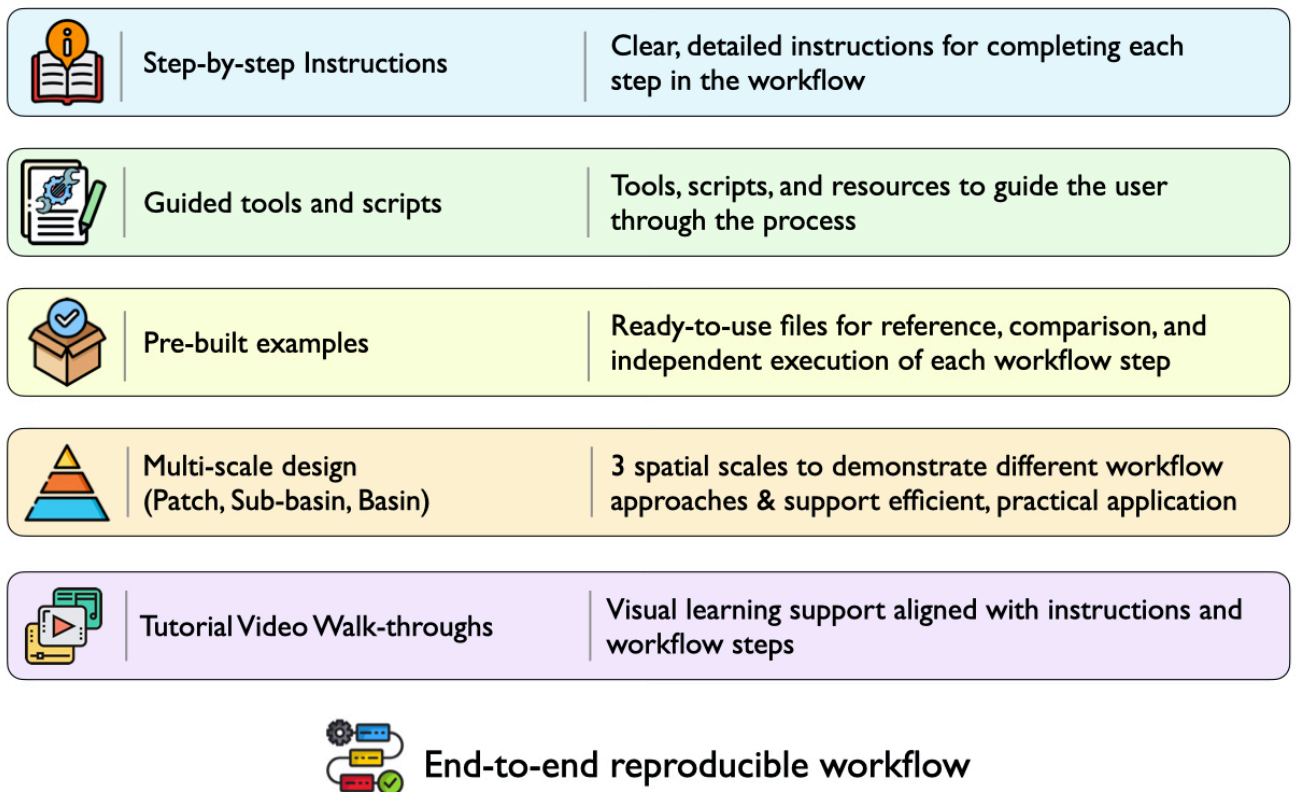


Figure 13.1: Instructional System Design

13.1 4.1 Study Site Overview

- Location
- Characteristics

13.2 4.2 Dataset Contents

- Raw Data: spatial (and workflow file to generate rasters), formatted climate and observational data
- Preprocessing data: Workflow script, all prerequisite files necessary for generating Worldfile/Flowtable, and completed Worldfiles and Flowtables (for a patch, hillslope, and full basin)
- Simulation files: Ready-to-run workflow scripts with functions to generate all necessary instruction to run the model for different strategies (i.e., single run, multi-run, for calibration & sensitivity analysis)
- Reference outputs: Sample model results
- How to get it

13.3 4.3 How to use this data

- Contextual understanding
- Performance Evaluation - verification for user generated files
- Troubleshooting
- Modular learning - start at any step in the manual

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